

IVAN RODRIGUEZ

AI Software Graphics Engineer — Full-stack engineering, machine learning, tuning, 3D and 2D graphics, and mathematical modeling
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PERSONAL PROFILE

- Full-stack developer with proven expertise including data science, and mathematical modeling.
- Proven ability to author and develop high-tech, client-requested services for platforms.
- Languages: Spanish Native, English C2, Japanese N4
- Visa Status: 1G Visa, Mexican Passport

AI & MACHINE LEARNING

Explainable AI for Black-Box Models | M.Sc. research, National College of Ireland

Built visual explanation tooling for CNN, deep learning, and deep Q-learning models using LIME, SHAP, and Grad-CAM, producing per-prediction attribution maps for models with no native interpretability. Python, TensorFlow, Imagemagick.

Retinopathy Classification | M.Sc. research

Novel data augmentation approach using CNN with ReLU, max pooling, and dropout; orchestrated with Dagster. Python, TensorFlow.

Low-Data Price Prediction | Odetta

Derived a market data augmentation technique from the Central Limit Theorem and Fourier analysis, increasing price prediction accuracy on sparse entries by 70%.

PROFESSIONAL EXPERIENCE

April 2026 - Present | Craftyamigo, CA | Software Engineer - Graphics and 3D Pipeline

Implemented an automated acceptance, 3D processing, and AI pipeline for parts, orchestrator with LangGraph.

- Combine geometric algorithms with AI calls for automated assessment of parts orchestrated by LangGraph.
- Develop a prototype MCP server for AI Blender automated usage.
- Migrate system from Blender to ThreeJS removing the dependence on processing on an application.
- Create a review platform giving access to other team members, increasing the approval of new parts from 20 a day to 100 per person.
- Collaborate with a tight team of 4 people for a fast paced improvement of the platform.
- Operate a shared remote build environment (Tailscale, tmux sessions) supporting 4 members across Guadalajara and California.

January 2023 - May 2024 | Glezztech, TX | Software Engineer (Remote)

Implemented and automated testing for a classification platform software that differentiates between human and machine users, similar to ReCaptcha.

- Ensure complete test case coverage across the whole project using Bash script for automatic detection and Jest for test cases.
- Develop custom Shell tools for automatic signaling and creation of missing test cases.
- Implementation of the application using React Native, Typescript and React Hooks.

March 2022 - January 2023 | Odetta, CDMX | Data Scientist (Remote)

Platform development and assist CTO with Mathematical and Statistical insight on the Pricing Model used by the company.

- Developed a market data augmentation technique using CLT and Fourier analysis that increased the price prediction accuracy of low-data entries by 70%.
- Boosted and merged the finite state machine for transactions by tracking the rest of the site's internal transactions and simplifying the existing state machines resulting in a cleaner, direct, up-to-date, company dashboard.

December 2018 - March 2022 | Envato, GDL | Software Engineer

Co-author and develop several high-tech client-requested services to further increase the capabilities of the overall platform for both users and clients.

- Co-created an Experiment Manager, enabling the engineering department to deploy statistically correct multivariate experiments and AB testing using a single experiment ID per client/user. This project allowed the company to eliminate the use of Google Optimize and run simultaneous AB and multivariate experiments on the platform, providing enriched analytics capabilities compared to the limited AB testing offered by Google Optimize.
- Engaged in the ML Background Removal project initiative to explore mathematical insights and image processing techniques.

- Deployed client-requested features, including “Export to PDF” and “Resize/Crop Image”, enhancing the platform experience for over 200,000 subscribers.
- Accomplished code analysis, optimization, cleanup, and documentation tasks for the Graphic Processing Engine, resulting in the eradication of errors from new smart templates and a significant reduction in debugging time for existing errors, mainly confined to the DSL or edge cases, freeing a daily 1 hour of work from 3 different teams of 20 people.
- Co-created an automatic mesh warp feature using displacement maps, reducing the time for simple mockups from 1 hour to half or less for a team of around 20 people.
- Received recognition for further improvements made on the Graphic Processing Engine and documentation, facilitating seamless enhancements, knowledge transfer, reducing time spent by 40%.

January 2016 - December 2018 | Placeit, GDL | Frontend Engineer

Automated Photoshop and later replaced Photoshop with a custom-built client-side graphic-rendering engine.

- Collaborated in developing a Multi-layer Graphic Processing Engine, replacing Photoshop and saving significant costs and processing time. Integrated reversed-engineered visual effects modules seamlessly and adapted API functions, porting thousands of templates with minimal platform usability disruption.
- Analyzed comparative performance between Photoshop and the new Engine, resulting in reduced processing times, decreased server needs, and lower costs.
- Researched, documented, and experimented on several visual design axioms, resulting in the implementation of automatically rebalancing smart templates which became a key product.
- Designed a scripting DSL and API, adopted across 3 departments of 12 people each, to create smart templates. The API provided the smart templates with independence of processing engine, improving functionality and versatility.

EDUCATION

September 2024 - September 2025 | M. Sc. in AI | National College of Ireland

Data Analysis for AI, Machine Learning, Emerging AI Technologies, Intelligent Agents, AI-Driven Decision Making, Data Governance

Project: Retinopathy Classifier

- Higher accuracy with novel approach for data augmentation
- Python, JupyterNotebook, TensorFlow, Dagster, CNN ReLu with MaxPooling and dropout layers

Project: Explainable AI Tools

- Visual Explanations for Black Box AI models such as CNN, DL, DQL
- Python, TensorFlow, LIME, SHAP, GradCam, Imagemagick

2009 - 2015 | B. Sc. in Computer Systems | ITESM | Result: 8.5 - Equivalent: First Class Honors

Compiler design, Computer Graphics, Quantitative Methods and Simulation, Intelligent Systems, Web App Development, Software Design and Architecture

TECHNICAL SKILLS

LANGUAGES: Python, JavaScript, TypeScript, Go, C, C++, C#, Ruby, Perl, PHP, Bash, Assembly, SQL

BACKEND & DATA STORES: Express, Rails, MySQL, PostgreSQL, MongoDB

ML & DATA: TensorFlow, scikit-learn, Pandas, NumPy, Seaborn, Matplotlib, LIME, SHAP, Grad-CAM, Dagster, LangGraph

TESTING & TOOLING: Jest, Selenium, Postman, Parcel, NPM, Conda, Git, VSCode, Dot

FRONTEND & GRAPHICS: ThreeJS, D3, PixiJS, GSAP, Lottie, React, React Native, Angular, NodeJS, HTML5, CSS, ES6, SVG, XML, JSON, Bootstrap

INFRASTRUCTURE & DEVOPS: Docker, AWS, Grafana, Tailscale, Jupyter Notebooks, Google Colab, RStudio

INTERESTS AND ACHIEVEMENTS

- Level 2 Member of the Persuasive Influence path @ **Toastmasters International**
- Attended a conference on automating data pipeline with Dynamic tables @ **Snowflake Dublin**
- Javascript Certification @ **Hackerrank**
- NASA Space Apps GDL Hackathon @ **ITESM GDL**
- Volunteer Social Service @ **Unidos GDL**
- Baking cookies and building mobile apps